How to Assemble the Magnet

The first step in the magnet assembly is to prepare all the tools and parts required for the process. We made a tools and parts list. This guides participants on how to track what tools are present, parts added to the assembly, and after disassembly, if all parts are added back to the case.
Parts needed for the Magnet assembly included Rings **#321, #251(2), #288(2), #42(2),** and their corresponding lids, 54-M3 heat set inserts, 54-M3 nylon screws to fasten the lids onto the rings, 48 ring spacers, 8-M4 dome-shaped nuts, 16 washers, 110 neodymium magnets, and 1 reference magnet.
Tools include 4 screw drivers, 2 spanners, 1 compass, 4 permanent markers, 1 vernier caliper, and 1 plastic ruler.
Since we are not permitted to use soldering guns in conference halls, we solder the heat-set inserts into the rings before the build sessions. Here, we install the Heat-Set Inserts into the rings by plugging the soldering gun into the power supply and allowing the nozzle to reach a certain high temperature for installing brass heat-set inserts. This is shown in Figure 1 below. We position each brass heat-set insert into each designated hole located along the inner-most circular region of the rings (i.e., the holes through which nylon/ brass nuts pass while covering the rings with lids. During this process, we apply controlled heat and gentle pressure to seat each insert flush with the ring surface, ensuring proper alignment and full embedding within the material. After installation, place the rings on a flat surface like a table top to allow the inserts to cool and solidify completely before going onto the next step.

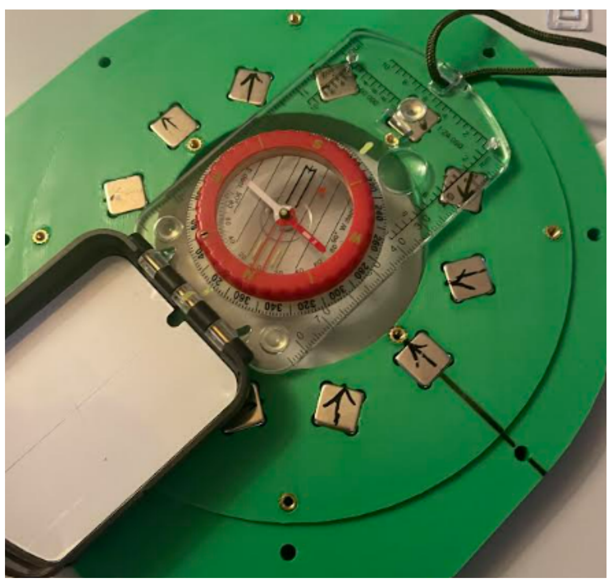

Figure 1: Showing Ring 42 with the heat inserts installed.

Magnet assembly

Step 1- Lay out all the parts on the table

First, all components are removed from the case and systematically arranged on a flat table to allow clear visualization, inspection, and preparation for subsequent assembly or analysis.

Step 2 - Arrangement of neodymium magnets.

Then, the neodymium magnets are removed from their respective storage boxes. During this process, they are arranged vertically to ensure organized placement and facilitate subsequent identification.

Step 3 - Determination of the reference magnet polarity.

A magnet is then selected to serve as a reference magnet. A Compass is calibrated so that the Compass needle at the center of the compass points in the South direction of the Earth's geographical magnetic field. That was identified as the compass's north pole, while the opposite end corresponded to the South Pole.

The Compass is then brought close to the selected reference magnet without allowing contact. The face of the magnet that attracts the compass's North Pole corresponds to the magnet's South Pole, and the opposite is the North Pole. This face corresponds to the magnet's South Pole. This reference magnet is then marked off by drawing an arrow indicating the magnet's North Pole direction, using a permanent marker. A small piece of cardboard is placed on the opposite face of the reference magnet using masking tape to provide physical separation and prevent unintended magnetic interaction during subsequent polarity identification of the remaining magnets.

Step 4 - Polarity Identification and marking of remaining magnets.

A row of magnets is brought into proximity with the reference magnet while maintaining controlled and careful handling to avoid sudden contact and knocking that might lead to cracking of the magnets. This is shown in Figure 2 below. The face of each magnet that is attracted to the south pole of the reference magnet is identified and later marked off as the North pole, so the arrow faces in that direction for all those magnets. This procedure is symmetrically repeated for all remaining magnets until the polarity of every magnet has been identified and marked clearly, using permanent markers.

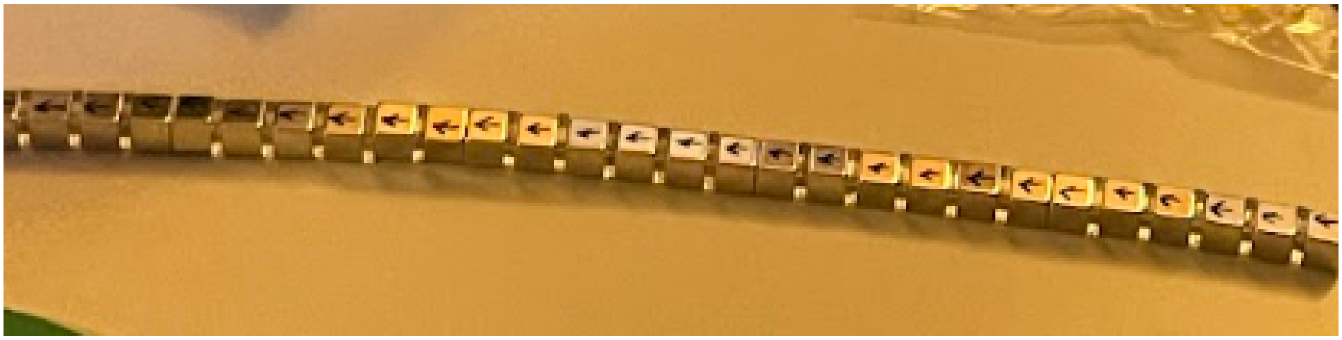

Figure 2: Showing a row of magnets laid out with marked polarities.

Step 5 - Placing magnets in their respective magnet slots of the rings.

First, we check each neodymium magnet for any dents or cracks that might lower the magnetic field strength contribution of each magnet. One could also place 1 magnet close to the Gauss meter probe to measure the magnetic field strength, which should be >>>>1T for the N48 Neodymium individual magnets. Each ring is then picked out, and magnets with marked polarities are inserted in its slots, starting with the magnet slot oriented at 0 degrees from the horizontal axis. While inserting the magnets, the arrow points in the same direction as the small circle on the magnet slot that was meant to guide a person who is placing magnets in the ring. Figure 3 shows the alignment of magnets in ring 42.

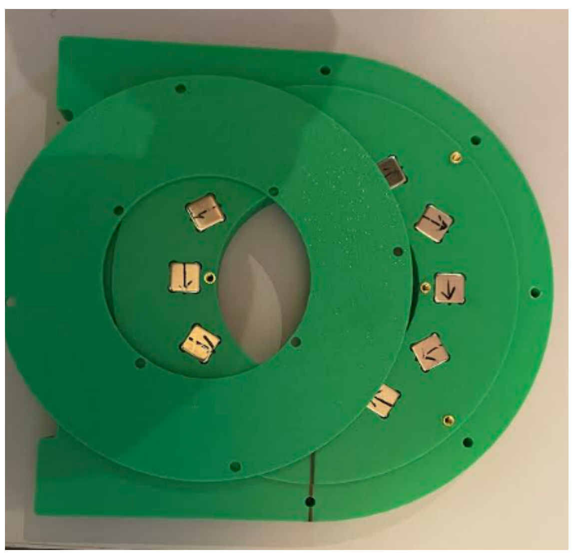

Figure 3: Showing alignment of Neodymium magnets in Ring #42.

Step 6 - Check Magnet orientation if done properly.

Once a participant has finished placing the magnets into a ring, another participant checks it to ensure that the magnets are inserted inside the ring slots in the appropriate order. One more essential check is done using a compass (Figure 4), in which case the compass is first rotated at the center of the ring to ensure that in all directions the compass is twisted, the Bo field is in one specific direction. When the same compass is moved around the external area of the ring, the compass needles rotate indefinitely, which confirms the Halbach array principle, with a magnetic field Bo in one direction at the center of the ring, and fields that are in all directions at the exterior of the ring. This is done for all rings.

Figure 4: Showing a participant using a compass to check if the magnets were properly aligned in the ring.

Step 7 - Cover the rings with the corresponding lids

Nylon screws/ brass screws and M3 screw drivers are used to fasten the ring lids onto the corresponding rings. The lids are made to be very tight (Figure 5).

Figure 5: Showing rings with magnets in their slots and lids tightened onto them using lids.

Step 8 - Make the Halbach array assembly.

Starting with brass rods, 8 dome-shaped nuts are placed at the end of each rod, then washers are added on top of each dome-shaped nut. Ring 42 was slid into the rods while pushing gently to the end of the rods, until it sits on the washers. Ring spacers are then added next in the 8 brass rods, and then the next Ring 252 is added. Spacers are added again, and Ring 288 is added. More spacers followed by Ring 321 are added. Spacers are added and the pattern repeated for rings #288, #252, and #42. Lastly, 8 washers are added to the brass rods, and the assembly is completed with brass nuts, using spanners. A vernier caliper was used to ensure that the distance between 2 rings is 12mm.

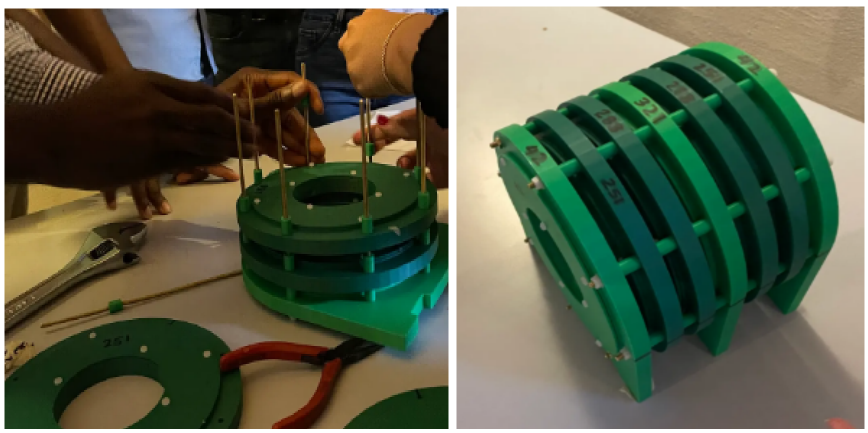

Figure 6: Showing the ring assembly process and the result of the Halbach array magnet after assembly.